B. Outstanding Impressionist

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

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If it was so, then let's make it a deal...

— MayDay, *Gentleness*

Even after copying the paintings from famous artists for ten years, unfortunately, Eric is still unable to become a skillful impressionist painter. He wants to forget something, but the white bear phenomenon just keeps hanging over him.

Eric still remembers n pieces of impressions in the form of an integer array. He records them as $w_1, w_2, \ldots, w_n$. However, he has a poor memory of the impressions. For each $1 \le i \le n$, he can only remember that $l_i \le w_i \le r_i$.

Eric believes that impression i is *unique* if and only if there exists a possible array $w_1, w_2, \ldots, w_n$ such that $w_i \ne w_j$ holds for all $1 \le j \le n$ with $j \ne i$.

Please help Eric determine whether impression i is *unique* for every $1 \le i \le n$, **independently** for each i. Perhaps your judgment can help rewrite the final story.

Input

Each test contains multiple test cases. The first line of the input contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. The description of test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$) — the number of impressions.

Then n lines follow, the i-th containing two integers l_i and r_i ($1 \le l_i \le r_i \le 2 \cdot n$) — the minimum possible value and the maximum possible value of w_i.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a binary string s of length n: for each $1 \le i \le n$, if impression i is *unique*, s_i=1; otherwise, s_i=0. Do **not** output spaces.

Example

Input

```
5
2
1 1
1 1
4
1 3
1 3
1 3
1 3
6
3 6
2 2
1 2
1 1
3 4
2 2
7
3 4
4 4
4 4
1 3
2 5
1 4
2 2
3
4 5
4 4
5 5
```

Output

```
00
1111
100110
1001111
011
```

Note

In the first test case, the only possible array w is [1, 1], making neither impression 1 nor 2 *unique* (since $w_1 = w_2$).

In the second test case, all impressions can be made *unique*:

- For $i = 1$, we can set w to [1, 3, 2, 3], in which $w_1 \neq w_2$, $w_1 \neq w_3$, and $w_1 \neq w_4$;
- For $i = 2$, we can set w to [2, 3, 1, 2], in which $w_2 \neq w_1$, $w_2 \neq w_3$, and $w_2 \neq w_4$;
- For $i = 3$, we can set w to [1, 1, 3, 1];
- For $i = 4$, we can set w to [2, 3, 3, 1].

In the third test case, for $i = 4$, we can set w to [3, 2, 2, 1, 3, 2]. Thus, impression 4 is *unique*.